

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.5: Area of Part of a Circle — Teacher Reference (Pre-filled)

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.5: Area of Part of a Circle
Driving Question	DRIVING QUESTION: How can we calculate the exact area of any wedge-shaped or segment-shaped piece of a circle — like a watermelon slice at Wakulima Market — so that we can solve fair-sharing and design problems in real life?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon Launch — Sector Intuition at Wakulima Market	Learners observed four watermelon-slice sectors — all with radius 7 cm but with central angles of 30°, 90°, 120°, and 180° — drawn on centimetre-square grid paper. By counting squares, they found that the estimated area increases as the central angle increases, and that Sector D (180°) has approximately twice the area of Sector B (90°), mirroring the doubling of the angle.	The area of a sector-shaped piece of a circle is NOT fixed by the radius alone — it changes dramatically with the central angle. When the radius is the same, doubling the central angle approximately doubles the area. Two variables control the size of a watermelon slice: the radius (r) and the central angle (θ). Prior methods for finding area (rectangles, triangles) cannot handle the curved boundary of a sector.	This begins to explain why Mama Wanjiku's customers receive such different amounts of watermelon even though every piece was cut from the same melon: the vendor changed the angle of each cut, and because area is proportional to the central angle, a small change in angle produces a large difference in the amount of fruit. A 30° slice contains only one-sixth as much watermelon as the 180° slice, even though both pieces look 'like a slice' and share the same radius. The exact formula — which will be derived in Lesson 2 — is needed to calculate fair shares precisely.
Lesson 2 Deriving the Sector Area Formula — From Fraction of a Circle to a Precise Formula	Learners physically tiled pre-cut sector cut-outs onto a full paper circle of radius 14 cm and observed that four 90° sectors, eight 45° sectors, or approximately 2.67 of the 135° sectors reconstitute the full circle — confirming that each sector is a measurable fraction ($\theta/360^\circ$) of the whole.	The area of a sector equals the fraction of the full circle it occupies multiplied by the full circle area: $\text{Area} = (\theta/360^\circ) \times \pi r^2$. For Mama Wanjiku's watermelon ($r = 14$ cm), the 45° slice ≈ 76.99 cm ² , the 90° slice ≈ 153.99 cm ² , and the 135° slice ≈ 230.98 cm ² — confirming that the 135° slice is exactly three times larger than the 45° slice.	The fair-sharing dispute at Wakulima Market can now be resolved mathematically: Mama Wanjiku's slices are provably unequal because they have different central angles. The sector area formula gives every customer an objective tool to verify the size of their slice in cm ² , directly answering the driving question for wedge-shaped pieces. The remaining puzzle is whether the same approach works for a segment (a slice bounded by a straight

			chord rather than two radii).
Lesson 3 Investigating Sector Area with Cut-Outs	Learners constructed paper cut-out sectors of a circle with varying central angles (30°, 60°, 90°, 120°, 180°) at a fixed radius of 7 cm, and sectors at a fixed angle of 60° with varying radii (4 cm, 7 cm, 10 cm). They measured, recorded data in tables, and identified patterns — that area scales linearly with central angle and as the square of the radius.	The area of a sector of a circle is given by $A = (\theta/360) \times \pi r^2$, where θ is the central angle in degrees and r is the radius. Doubling the angle doubles the area; doubling the radius quadruples the area. Learners applied this formula to calculate the areas of two Wakulima Market watermelon wedges ($\theta = 90^\circ$ and $\theta = 180^\circ$, $r = 7$ cm) and determined the vendor's equal-price claim was mathematically unfair.	The sector area formula (derived in Lesson 2) comes from the fact that a sector is a fraction ($\theta/360$) of the full circle. Because the full circle has area πr^2 , any sector has area $(\theta/360) \times \pi r^2$. This explains why the vendor's two wedge-cuts at different angles produce measurably different areas even though both come from the same melon — changing only the angle of the cut directly and proportionally changes the area of the piece.
Lesson 4 Cutting Watermelon Fairly: Applying the Area of a Sector Formula	Learners cut out five sectors of angles 30°, 60°, 90°, 120°, and 180° from a printed circle of radius 7 cm, arranged them in order, measured their angles and radii, and recorded the full circle area ($\pi r^2 \approx 154$ cm ²) as the baseline.	The area of a sector is a fraction of the full circle area determined by the central angle: Area of sector = $(\theta/360) \times \pi r^2$. A 90° sector contains exactly one quarter of the full circle area; a 180° sector contains exactly half. Changing only the angle — while keeping the radius constant — changes the area proportionally.	Learners applied the formula to two new Kenyan contexts (a 45° ugali cake slice with $r = 14$ cm, and a door sweep of 120° with $r = 80$ cm in Kisumu), updated their cumulative watermelon model with the formula and labelled sector areas, and added the connecting sentence: 'This formula tells the mama mboga exactly how much watermelon any wedge contains so every customer is charged fairly by area.'
Lesson 5 Lesson 5 — From Wedge to Segment: Deriving and Applying the Area of a Circular Segment	Learners observed that when a chord is drawn across a sector of Mama Wanjiku's 14 cm radius watermelon, the sector is divided into two distinct regions — an isosceles triangle (formed by the two radii and the chord) and a curved segment (the region between the chord and the arc). They saw physically and numerically that Segment = Sector – Triangle, and verified this with the specific case of a 60° sector.	The area of a circular segment is found by subtracting the area of the isosceles triangle from the area of the sector that contains it: Area of segment = $(\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta$. The triangle's area is calculated using the two-sides-and-included-angle formula $\frac{1}{2}r^2 \sin \theta$, which connects to the triangle area work in Sub-Strand 2.4 Area of Polygons. A segment is bounded by one chord and one arc; a sector is bounded by two radii and one arc — these require different formulas.	The phenomenon involves two types of curved pieces: the pointed wedge (sector) and the flat-cut curved rind-piece (segment). Now that learners have both the sector formula (Lessons 2–4) and the segment formula (Lesson 5), they can calculate the exact area of ANY piece Mama Wanjiku cuts from her watermelon — whether she cuts along two radii (a sector) or along a chord (a segment). This means a customer at Wakulima Market can now verify mathematically, not just visually, whether their flat-cut curved slice is a fair share.
Lesson 6 Applying the Sector Area Formula: From Watermelon Wedges to Irrigation Arcs	Learners observed a teacher-led step-by-step recap of the derivation of the sector area formula $A = (\theta/360) \times \pi r^2$ from first principles, built on the Wakulima Market watermelon phenomenon. They also observed three real-life Kenyan sector contexts: a 120° watermelon slice ($r = 15$ cm), a school gate sweep ($r = 1.2$ m, $\theta =$	Learners learned that the area of any sector is a proportional fraction of the full circle's area, expressed as $A = (\theta/360) \times \pi r^2$. They learned that sector area scales linearly with angle θ (doubling the angle doubles the area) and quadratically with radius r (doubling the radius quadruples the area). They applied this formula to calculate areas	Learners explained in their cumulative model diagrams and DQB Formula Cards why changing only the angle of a cut at Wakulima Market changes the area of the watermelon wedge so dramatically: because sector area is directly proportional to the central angle ($A \propto \theta$), a vendor who cuts at 120° instead of 60° produces a slice

	75°), and a Naivasha centre-pivot irrigation arc ($r = 80$ m, $\theta = 270^\circ$). Learners observed peer pair presentations on mini-whiteboards and whole-class evaluation of solution accuracy.	in cm^2 , m^2 , and hectares across three graduated Kenyan contexts, and updated the class Driving Question Board with the formalised formula and new arising questions pointing toward applying sector and segment area together in Lesson 7.	exactly twice as large — resolving the customers' complaint with mathematical precision. Learners also explained the r^2 relationship in a written Scaling Note, and exit card responses documented both procedural fluency (correct substitution) and conceptual understanding (proportionality reasoning without calculation).
Lesson 7 Investigating Segment Area: From Sector to Slice — Unlocking the Curved Piece	Learners cut out a sector of radius 14 cm and 60° , separated it into a triangle (OAB) and a segment, and calculated each area. They also completed a class data table for angles 60° , 90° , and 120° , recording sector area, triangle area, and segment area for each.	The area of a segment = area of sector – area of triangle = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$. The triangle formed by two radii and a chord has area $\frac{1}{2}r^2\sin\theta$. A segment is always smaller than its corresponding sector because the triangle area is always positive.	For the mama mboga's watermelon at Wakulima Market: when a wedge is cut straight across a chord, the customer receives a segment — the rind-bounded piece between the chord (cut face) and the curved rind arc. Using the segment formula, the exact area of this piece in a 90° slice of radius 21 cm is 125.9 cm^2 . This means we can now price each piece fairly by area, not by visual estimate.
Lesson 8 Segment Area, Final Explanation & Fair-Share Verdict — Completing the Watermelon Puzzle	Learners physically cut the triangle (formed by two radii and the chord) from their sector cut-outs to isolate the curved segment, measured dimensions, and completed a Group Data Table recording sector area, triangle area ($\frac{1}{2}r^2\sin\theta$), and segment area for four different central angles (40° , 65° , 90° , 120°) using $r = 18$ cm — modelling a new Wakulima Market watermelon.	Learners derived and applied the formula Area of segment = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$, verified it numerically against their cut-and-subtract results, used it to adjudicate the Customer Complaint Card (calculating which of Mama Wanjiku's four customers received the largest curved segment), and connected both sector and segment formulae to real-life fair-sharing and design contexts in Kenya.	Learners produced a complete, evidence-based Final Explanation of the driving question — articulating that the area of any wedge (sector) or curved-slice (segment) of a circle is determined precisely by the central angle θ and radius r through their respective formulae, not by visual appearance alone. They revised their cumulative circle-area model (annotating all three formulae onto their Lesson 1 naïve drawing), completed the Driving Question Board audit, and gave Mama Wanjiku a mathematically justified recommendation for ensuring every customer at Wakulima Market receives a truly fair share.

ATTRIBUTION AND LICENSING

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